

**GOVERNMENT ARTS AND SCIENCE COLLEGE,
ALANGUDI**

College code:BDU752

DEPARTMENT OF COMPUTER SCIENCE

PROJECT REPORT

FITBUDDY

SUBMITTED BY

TEAM III

NAME	EMAIL ID	REG NO
Muthukumar A (Team Leader)	muthukumar.a1402@gamil.com	25057521802111019
Karthik M	kkarthi99773@gmail.com	25057521802111015
Ajithkumar S	ajithkumar1822008@gamil.com	25057521802111001
Deepak R	deepak10una@gamil.com	25057521802111006
Janarthanan S	Selvamani33821@gamil.com	25057521802111013

PHASE 1

**FitBuddy- AI Fitness Plan Gen erator using
Gemini Models**

Brainstorming and Ideation Phase

Introduction

Brainstorming and ideation is the initial phase of the FitBuddy project. The main purpose of this phase is to identify the problem that the project aims to solve, generate possible solutions, understand the role of Artificial Intelligence, and establish the basic concept of the proposed application.

The project **FitBuddy – AI Fitness Plan Generator** was conceptualized as an AI-powered web application that can generate personalized fitness plans according to the individual requirements of users.

The idea focuses on combining fitness planning with Generative Artificial Intelligence. Instead of providing the same workout routine to every user, FitBuddy uses user-provided information such as name, age, weight, fitness goal, and workout intensity to generate a customized plan.

The project documentation describes four major scenarios for FitBuddy:

1. A user enters personal fitness information and receives a personalized 7-day workout plan.
2. A user provides feedback about an existing plan and receives an updated plan.
3. A user receives a nutrition or recovery tip based on the selected fitness goal.
4. An administrator or coach can view registered users and their original and updated workout plans.

Problem Identification

People have different fitness goals, physical conditions, preferences, and workout intensities. A single fixed workout plan may therefore not be suitable for every individual.

For example, one person may want to focus on weight loss, while another may want to build muscle. Similarly, users may prefer low-, medium-, or high-intensity workouts.

The brainstorming process identified the need for a system that could take basic user requirements and generate a structured fitness plan automatically.

Traditional fitness planning may also require users to manually search for exercises and organize them into a weekly schedule. This can be time-consuming, especially for beginners.

Therefore, the project idea was formed around the following problem:

How can Artificial Intelligence be used to automatically generate a personalized and structured fitness plan based on a user's requirements?

Identification of the Project Opportunity

The development team identified Generative AI as an opportunity to improve personalized fitness planning.

Instead of simply displaying predefined workout information, an AI-based system can process the user's input and generate a response based on the provided requirements.

The FitBuddy concept therefore combines:

- User information
- Fitness goals
- Workout intensity
- Generative AI
- Personalized workout generation
- Nutrition/recovery guidance
- User feedback
- Database storage

The uploaded project document specifically describes FitBuddy as an AI-powered application designed to generate personalized workout plans and adapt them based on user feedback.

Idea Generation

During the ideation process, several possible approaches to fitness planning were considered.

Idea 1 – Fixed Workout Library

One possible approach was to create a collection of predefined workout routines.

However, this approach would provide limited personalization because users would mainly select from existing plans.

Idea 2 – Rule-Based Workout Generator

Another approach was to create a rule-based system.

For example:

- If goal = weight loss → provide cardio exercises.
- If goal = muscle gain → provide strength exercises.
- If intensity = low → provide beginner-level exercises.
- If intensity = high → provide advanced exercises.

Although this approach could provide basic customization, it would require many manually created rules.

Idea 3 – AI-Based Workout Generator

The final concept focused on using Generative AI to create personalized workout plans.

The AI can receive information about the user's goal and intensity and generate a structured plan.

This approach became the foundation of FitBuddy.

FitBuddy Concept

The final idea was to develop a web-based application called:

“FitBuddy – AI Fitness Plan Generator”

The application allows a user to enter information such as:

- Name
- User ID
- Age
- Weight
- Fitness Goal
- Workout Intensity

The project document specifies fitness goals such as weight loss and flexibility and workout intensity levels such as Low, Medium, and High.

After the information is submitted, the system processes the information and generates a personalized workout plan.

The resulting plan can contain:

- Daily workout focus
- Exercises
- Sets
- Repetitions
- Duration
- Rest intervals
- Warm-up guidance
- Cool-down or recovery guidance

Role of Generative AI

Generative AI became the central technological idea of the FitBuddy project.

The project documentation records experimentation with several Hugging Face models, including:

- T5
- BART
- DistilBERT
- Bloom

The documented project ultimately selected Google Gemini models for the production implementation.

According to the project documentation:

Gemini 1.5 Pro

The documented implementation used Gemini 1.5 Pro for:

- Generating structured 7-day workout plans
- Updating workout plans according to user feedback

Gemini Flash

The documented implementation used Gemini Flash for:

- Generating concise nutrition or recovery tips

The project documentation gives the reason for the selection as the API-based integration, ability to understand structured formatting, minimal prompt tuning, and suitability for feedback-based revisions.

Note: These model names describe the implementation documented in the supplied project report.

Brainstorming Process

The FitBuddy brainstorming process can be represented as follows:

Fitness Problem



Identify User Needs



Generate Possible Solutions



Evaluate Rule-Based and AI-Based Approaches



Select Generative AI Approach



Define FitBuddy Concept



Identify Core Features



Prepare for Requirement Analysis

Figure 1 – Brainstorming and Ideation Flow

Illustrative diagram for documentation

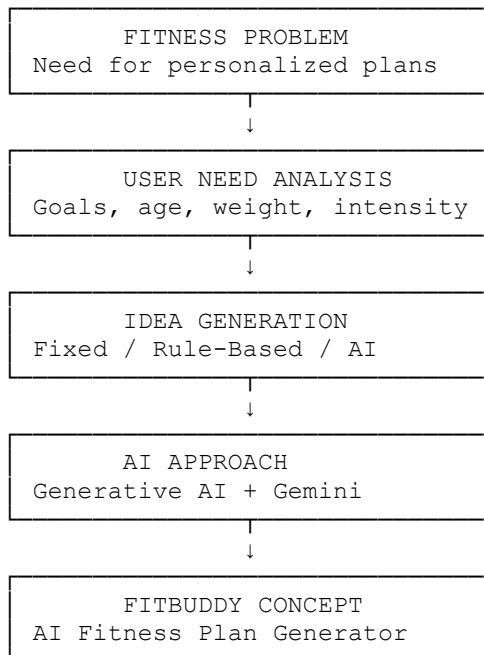


Figure 1: FitBuddy Brainstorming and Ideation Process

Core Features Identified During Ideation

The brainstorming phase identified the major functionalities that would later become system requirements.

Personalized Workout Generation

The primary feature is the generation of a personalized workout plan.

The system uses user information and fitness requirements to generate a structured 7-day workout schedule.

Nutrition and Recovery Tip

FitBuddy also provides a concise nutrition or recovery suggestion related to the user's fitness goal.

The project documentation gives examples such as hydration advice, protein suggestions, and general recovery practices.

Feedback-Based Plan Updating

The brainstorming phase also identified that a fitness plan should be adaptable.

Users can provide feedback such as:

- “Add yoga”
- “Include more cardio”
- Request additional rest
- Request changes to the workout

The documented system passes the original plan and user feedback to the AI model to generate an updated plan.

User and Plan Storage

The system also stores user information and generated plans.

The documented implementation uses SQLite with SQLAlchemy for persistence and keeps original and updated plans separately.

Administrative View

The project also includes a view for administrators or coaches.

This allows them to see registered users and their workout plans, including original and updated versions.

Figure 2 – Main FitBuddy Concept

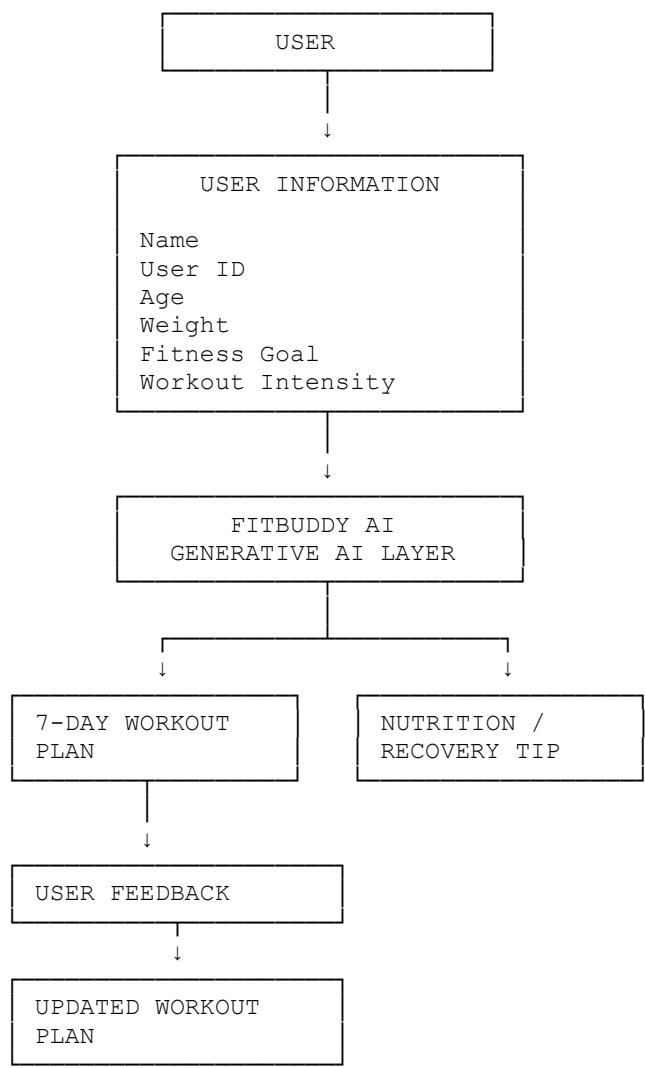


Figure 2: High-Level FitBuddy Concept

User Scenarios Identified

The ideation phase also defined the major scenarios that the application should support.

Scenario 1 – Generate a Workout Plan

A user provides fitness information and selects a goal and workout intensity.

The system processes the information and generates a personalized 7-day workout schedule.

Scenario 2 – Modify an Existing Plan

A user provides feedback about an existing plan.

The system uses the feedback to generate a revised workout plan.

Scenario 3 – Obtain Nutrition Guidance

A user requests a nutrition or recovery tip related to the selected fitness goal.

The application provides a concise suggestion.

Scenario 4 – Administrative Monitoring

An administrator or coach accesses the user information and plans.

The system displays the original and updated plans for registered users.

Figure 3 – User Interaction Concept

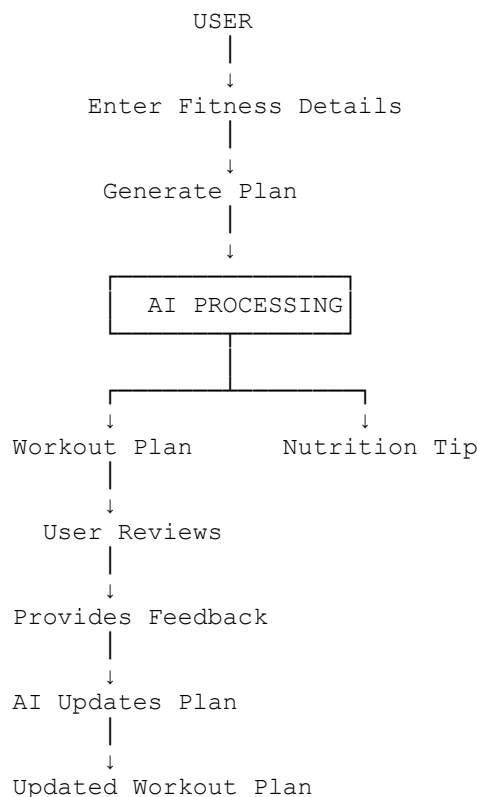


Figure 3: FitBuddy User Interaction Flow

Initial Technical Idea

Once the main concept was identified, the team considered the technologies required to implement the application.

The project documentation identifies the following major technologies:

Component	Technology
Programming Language	Python
Backend Framework	FastAPI
AI	Google Gemini
Database	SQLite
ORM	SQLAlchemy
Frontend	HTML/CSS
Template Engine	Jinja2
Server	Uvicorn

The architecture described in the project uses HTML and Jinja2 for the frontend, FastAPI for backend processing, Google Gemini for AI functions, and SQLite with SQLAlchemy for persistence.

Initial Architecture Concept

The brainstorming phase established a basic understanding of how the major components would interact.

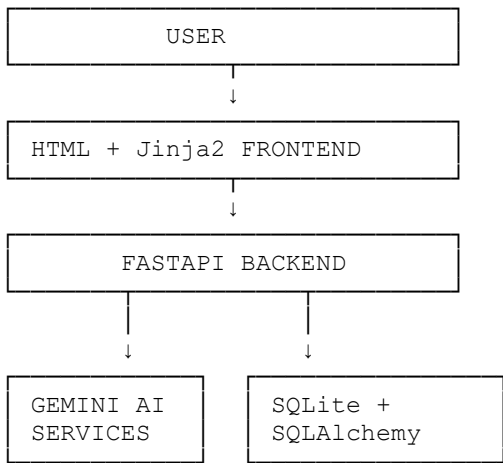


Figure 4: Initial FitBuddy Technical Concept

The detailed architecture was subsequently developed in the project-design stage. The source project describes the application as a modular FastAPI-based architecture with frontend, backend, AI, and database layers.

Expected Benefits of the Idea

The FitBuddy concept was designed around several expected benefits.

Personalization

The application uses user-specific information rather than providing one fixed workout plan.

Automation

Generative AI reduces the amount of manual effort required to create structured plans.

Adaptability

The feedback mechanism allows the generated plan to be revised.

Accessibility

The web interface allows users to access the application through a browser.

Structured Planning

The generated workout plan is organized into a day-wise schedule.

Fitness-Oriented Guidance

The system also provides nutrition and recovery-related suggestions.

Limitations Identified During Ideation

The brainstorming phase also needs to recognize limitations.

The AI-generated plan depends on the information provided by the user and the behavior of the selected AI model.

The application is intended as a fitness-planning project and should not be interpreted as a replacement for professional medical or clinical advice.

Other considerations include:

- Accuracy of user input

- Quality of AI-generated output
- API availability
- Internet connectivity for AI services
- API key configuration
- Appropriate handling of stored user information

These considerations can be addressed further during requirement analysis, development, and testing.

Ideation Outcome

At the end of Phase 1, the following concept was established:

FitBuddy is an AI-powered web application that accepts user fitness information and uses Generative AI to create a personalized 7-day workout plan, provide a nutrition/recovery tip, and revise the plan according to user feedback.

The project also includes persistence of user and plan information and an administrative view of users and plans.

Phase 1 Deliverables

The major outputs of the Brainstorming & Ideation Phase are:

1. Identification of the fitness-planning problem.
2. Identification of user needs.
3. Generation of possible solution approaches.
4. Selection of an AI-based approach.
5. Definition of the FitBuddy concept.
6. Identification of major user scenarios.
7. Identification of core application features.
8. Initial identification of the technology stack.
9. Initial architecture concept.
10. Foundation for the next Requirement Analysis Phase.

Phase 1 Summary

The Brainstorming and Ideation Phase established the foundation for the FitBuddy project.

The central idea was to combine **fitness planning and Generative AI** to produce personalized workout recommendations.

The project moved from identifying the general problem of personalized fitness planning to defining a concrete application concept.

The resulting FitBuddy concept includes:

**User Input → AI Processing → Personalized Workout Plan → Nutrition/Recovery Tip
→ User Feedback → Updated Plan**

This concept provides the foundation for the next phase, **Phase 2 – Requirement Analysis**, where the functional and non-functional requirements of FitBuddy can be formally identified and documented.

PHASE 1 CONCLUSION

The Brainstorming & Ideation Phase successfully establishes the basic direction of the FitBuddy – AI Fitness Plan Generator project. The project idea is centered on personalized fitness planning using Generative AI, supported by a web-based interface and persistent storage.

The ideation process identified the need for personalized workout generation, nutrition/recovery guidance, feedback-based plan modification, and administrative monitoring. The project documentation subsequently developed these ideas into a FastAPI, Gemini, SQLite/SQLAlchemy, HTML, and Jinja2 based application architecture.